

BRANDON CHEN

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EDUCATION

New York City College of Technology

Expected May 2027

B.S. in Mechanical Engineering | GPA: 3.60 / 4.00

- **Relevant Coursework:** Materials Science, Strength of Materials, Manufacturing Processes, Thermodynamics, Kinematics/Dynamics, Quality Control
- **Certification:** Basic CNC Programming and CAM Fundamentals - SolidProfessor / Mastercam Training (2025)

WORK EXPERIENCE

Amazon | Reliability & Maintenance Engineering

Summer 2026

Area Maintenance Manager Intern - LAS7, North Las Vegas, NV

- **Won the Top People's Choice Award at Amazon's Global AI Solutions Expo 2026 - one of four RME intern projects recognized out of 200+ submissions globally.**
- Led daily deep dives on top-ranked jam contributors from the Global OEE Faults dashboard, cross-referencing open work orders and PdM findings to separate controls faults from mechanical failures.
- Drove root cause analysis to corrective action on recurring failures, verifying each fix against a required 50% jam-frequency reduction over a 7-day window before closing the work order.
- Designed, built, and deployed two production dashboards reading live equipment state from Rockwell ControlLogix PLCs over EtherNet/IP, plus a material-handling SQL Server, the CMMS, and the ticket queue; re-validated at a second fulfillment center (SNV1) with a different equipment layout.
- Cut estimated shift-handover prep from ~30 minutes to ~5 by auto-generating the turnover artifact; reduced duplicate and miscoded tickets by pre-filling equipment ID, area, and live fault reason.

NYC Health + Hospitals / Woodhull

Jul - Aug 2023

Engineering Intern - Brooklyn, NY

- Performed preventative maintenance and corrective repairs on electromechanical medical systems (EKG, ultrasound, radiology); diagnosed faults using multimeters, circuit analysis, and structured troubleshooting, and documented root-cause findings and corrective actions in technical service reports.

PROJECTS

LAS7 Jam DPMO Standard Operating Procedure

Amazon Global Jam Program | 2026

- Authored the site SOP standardizing the daily deep dive on top-ranked jam contributors, root cause classification (controls vs. mechanical), JAM.REDUCT work-order creation and execution, and DPMO bridging across RME roles (Area Engineer, AMM, technicians).

MANE - Live Equipment Status Dashboard

Amazon RME | 2026

- Direct EtherNet/IP CIP reads (pycomm3) against Rockwell ControlLogix PLCs surfacing what is running, stopped, or jammed in seconds; per-controller tag map plus a single site-config file for portability across buildings.

AMM - Shift Command Dashboard

Amazon RME | 2026

- Single command page unifying downed equipment, work orders, PM compliance, condition-monitoring sensors, booked labor, tickets, and end-of-shift turnover; ~10,000 lines of Python (Flask/Jinja), PyInstaller-packaged, deployed on AWS EC2 behind nginx/TLS.

Arduino Robotics & IoT Systems Development

2026

- Programmed an Arduino UNO R4 Minima (C/C++) for sensor integration and closed-loop PWM motor control; built a smart-car navigation system with ultrasonic/IR obstacle detection and an ESP8266 WiFi module for IoT connectivity.

Advanced Parametric Design, CAM Programming & CNC Machining

2025

- Built fully constrained parametric CAD models with tolerance-driven design intent; generated contour, pocketing, drilling, and facing toolpaths in Mastercam, post-processed G-code, and machined parts to spec on a CNC mill.

Structural Design of Pedestrian Walkway Beam System

2024

- Modeled multi-load (pedestrian and wind) scenarios, constructed shear-force and bending-moment diagrams, performed stress and deflection analysis via elastic beam theory, and ran a steel-vs-concrete trade study.

Aircraft Structural Fabrication & Engine Maintenance

2019

- Fabricated composite aileron structures using precision riveting, epoxy bonding, welding, and heat treatment to aerospace tolerances; inspected reciprocating engines and pressure-tested fluid lines.

TECHNICAL SKILLS

- **Reliability & Analysis:** Root cause analysis, structured deep dives, jam DPMO & OEE analysis, PM compliance, predictive maintenance (PdM), LOTO
- **Controls & Automation:** Rockwell ControlLogix PLCs, ladder logic, EtherNet/IP (CIP), SCADA, SQL Server, CMMS / APM
- **Mechanical & Fabrication:** GD&T (ASME Y14.5), tolerance stack-up, stress/strain, shear & bending moment diagrams, beam deflection, riveting, welding, heat treatment, micrometers, torque wrenches
- **CAD/CAM & Programming:** SolidWorks, AutoCAD, Inventor, Mastercam, CNC, 3D printing; Python (Flask, pycomm3, pyodbc), C/C++, MATLAB, SQL
- **Robotics & Embedded:** Arduino UNO R4, Raspberry Pi, ESP8266 WiFi, ultrasonic/IR sensors, PWM control, H-bridge drivers